

CA 24K 0745

Q1) P1 & P2

P2 $\rightarrow$ 25% less time $\rightarrow$ 0.75 P1
20% more CPI

P1 clock freq = 1GHz

$$\text{CPU time} = \frac{IC \times CPI}{\text{Clock freq}}$$

$$\Rightarrow P1 = \frac{IC \times CPI}{1}, P2 = \frac{IC \times 1.2CPI}{x}$$

$$\frac{0.75(CPI)}{1} = \frac{1.2CPI}{x}$$

$$x = \frac{1.2}{0.75} = 1.6$$

$x = 1.6$ of P2 (clock freq)

Q2)

mttf = 2,000,000 single
2500 servers $\rightarrow$ 4 SSDs

$$a) \frac{10^9}{2000000} = 500$$

$$b) 2500 \times 4 = 100,000$$

$$fit = \frac{1}{\% \text{ mttf}} = \frac{1}{2000000} = 5 \times 10^{-7} \times 1000000 = 0.05\%$$

$$c) \frac{1}{fit} = \frac{1}{0.05} = 20$$

$$d) \text{Fit} = \frac{1}{1500,000} \times 100000 = 0.067$$

$$\frac{1}{0.067} = 15$$

$$e) \frac{15}{20} = 0.75 = 25\% \text{ reduction}$$

$$Q3) \Sigma (0.4 \times 1 + 0.25 \times 3 + 0.25 \times 3 + 0.1 \times 10) = 2.9$$

$$b) \frac{IC \times CPI}{ck} = \frac{30 \times 10^9 \times 2.9}{3 \times 10^9}$$

Q4) a) the period shows major improvements in processor architecture & semiconductor technology

- 1) ~~pipelining~~ pipelining was improved
- 2) superscalar processors were in use
- 3) cache improvements were made

b) The CPU made faster, created problems such as heat, cooling it became difficult this is known as power wall

c) The voltage could no longer be decreased as quickly as transistor dimensions so voltage stopped scaling as expected so power became a problem. The power of heat

would be too much.

- ii) it is due to power & heat
- chip temperature rises & heat can't be removed, the energy consumption increases

Q5)

- a) standardized benchmarks provide some workloads & testing conditions for different processors making performance comparisons fair.
- b) SPEC int. measures performance of integer based workloads
SPECfp : measures performance of floating point workloads
- c) different systems run the same standardized programs using defined rules of measurement methods, allowing fair comparisons
- d) Suite contains different workloads & gains a balanced view of processor
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performance.

e) processor performs better a completes
SPEC workloads faster

f) may not represent every real world
application so their result don't reflect
overall performance.